Dynamical Systems

Part II

Recorded from the lecturer in 2026-2027.

Dynamical systems – mathematical entities described by a set of equations for a set of variables w.r.t. a time-like variable.

State space or phase space.

∗ Continuous time ⇒ set of ODEs

Lotka–Volterra for rabbits (r) & sheep (s)

$$\dot{r} = r(a - br - cs), \qquad \dot{s} = s(d - er - fs), \qquad a, b, \ldots, f > 0.$$

$$\text{State space} \qquad [0, \infty) \times [0, \infty).$$

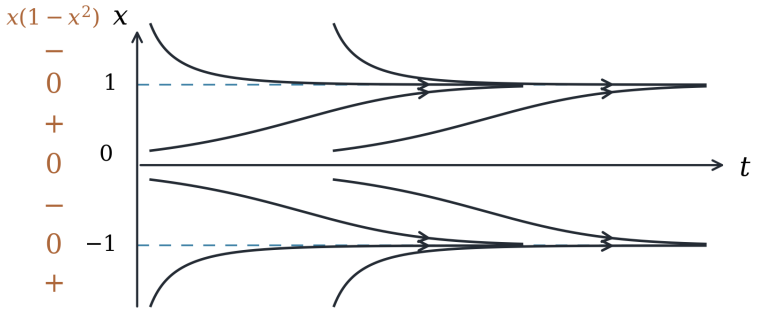
e.g. Logistic map $\qquad x_{n+1} = \mu x_n (1 - x_n), \quad 0 \le \mu \le 1$

Questions to address

- ∗ Long-term behavior of the system?
- ∗ Are there periodic solutions
- ∗ Stability of solutions, chaos
- ∗ Effect of system parameter variation

Example 1D system $\qquad \dot{x} = x(1 - x^2)$

$$\text{Solution} \qquad x(t) = \pm \left(1 + 2e^{2(t_0 - t)}\right)^{-1/2}$$



as $t \to \infty$

$x(0) > 0 \Rightarrow x(t) \to 1$

$x(0) < 0 \Rightarrow x(t) \to -1$

$x(0) = 0 \Rightarrow x(t) = 0$

Example 2D, Lotka–Volterra

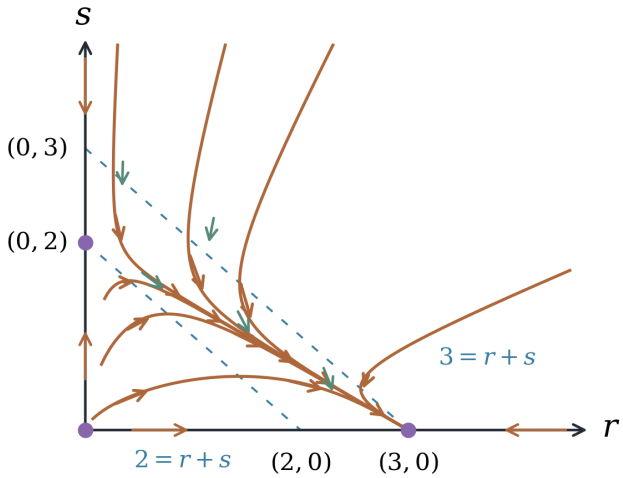
$$\dot{r} = r(3 - r - s)$$
$$\dot{s} = s(2 - r - s)$$

Can get $r(s) \Rightarrow t(s)$

∗ Find fixed points

$$\dot{r}, \dot{s} = 0 : \quad (0,0), (3,0), (0,2)$$

∗ Look at $\dot{r}, \dot{s}$ in state space



$$\text{As } t \to \infty \qquad (r,s) \to \begin{cases} (3,0), & r > 0, \\ (0,2), & r = 0,\ s > 0, \\ (0,0), & r = s = 0. \end{cases}$$

Example predator-prey

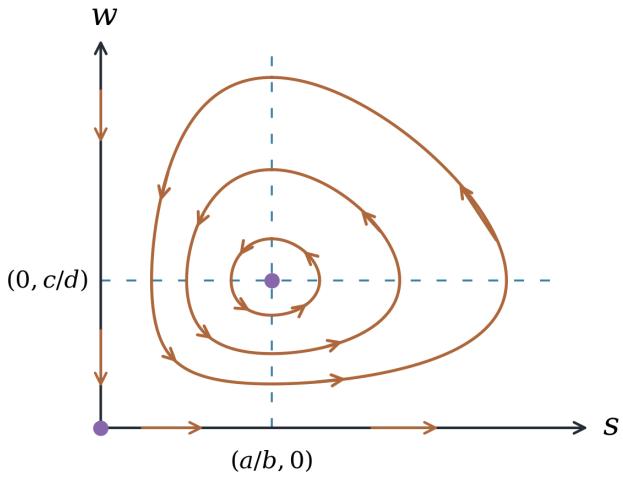
sheep (s) & wolves (w)

$$\dot{w} = w(-a + bs)$$
$$\dot{s} = s(c - dw)$$

$$a, \ldots, d > 0$$

Fixed points

$$(0,0), \quad (a/b, c/d)$$



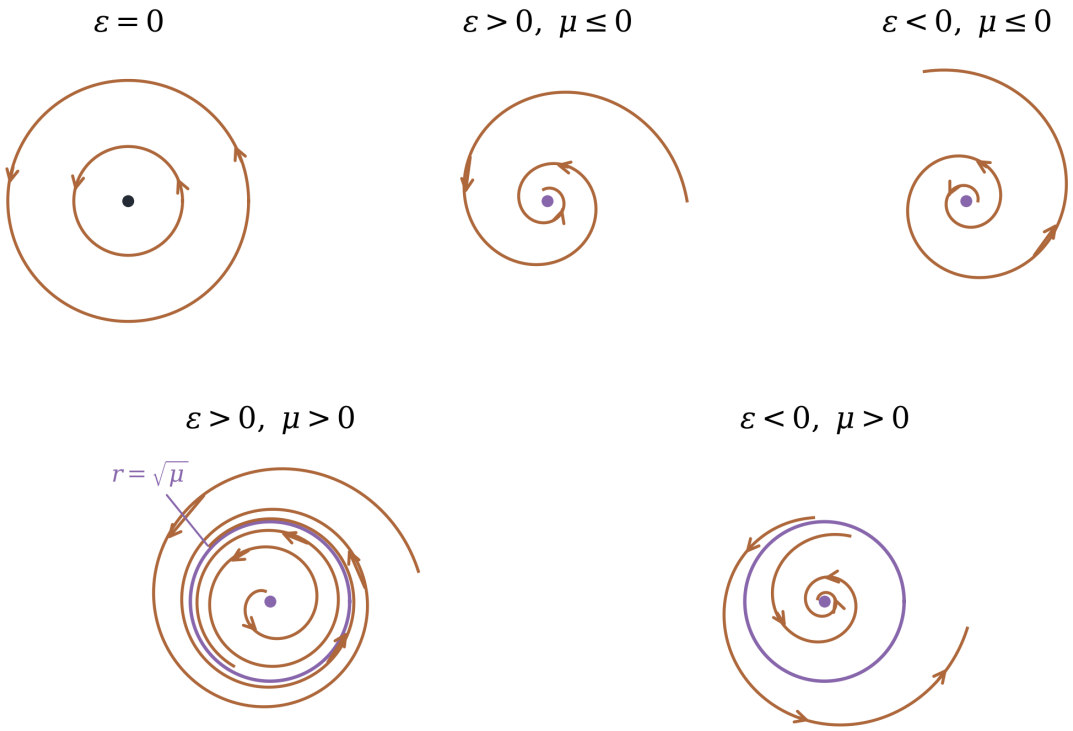
$$(s,w) \to \begin{cases} \text{periodic around } (a/b, c/d), & s > 0,\ w > 0, \\ (0,0), & s = 0,\ w \geq 0, \\ (\infty, 0), & s > 0,\ w = 0 \end{cases} \qquad \text{as } t \to \infty$$

Example

$$\dot{x} = -y + \varepsilon x(\mu - x^2 - y^2)$$
$$\dot{y} = \quad x + \varepsilon y(\mu - x^2 - y^2)$$

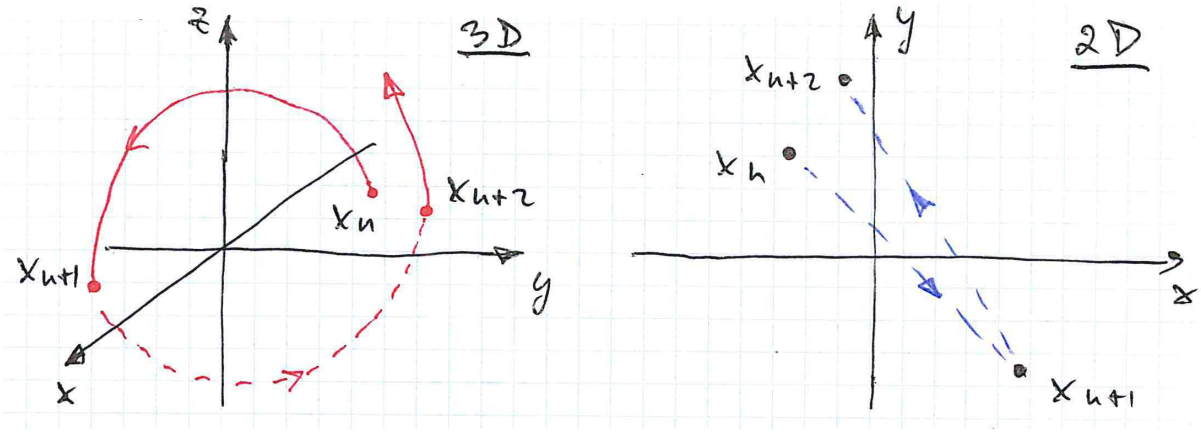
Switch to polars:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \operatorname{atan} \frac{y}{x},$$
$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \qquad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} \quad \Rightarrow$$
$$\underline{\dot{r} = \varepsilon r(\mu - r^2)}, \qquad \underline{\dot{\theta} = 1}$$



End of Lecture 1

Poincare map in 2D for a 3D dynamical system:



Can also consider maps as discretized ODEs.

Plan for the course

(3)	§1 :	Basic definitions	ODEs
(3)	§2 :	Fixed points	
(2)	§3 :	Stability	
(5)	§4 :	Periodic orbits	
(5)	§5 :	Bifurcations	
(3)	§6 :	Maps	Maps
(3)	§7 :	Chaos	

§1. Definitions

(1.1) Define a State Space (Phase Space) $E \subseteq \mathbb{R}^n$. The state of the system is described by $\bar{x} \in E$. The state evolves in time according to an ODE of the form

$$\dot{\bar{x}} = \frac{d\bar{x}}{dt} = \bar{f}(\bar{x}, t), \qquad \text{where } \bar{f} : E \times \mathbb{R} \to E \text{ is a vector field.}$$

If $\dfrac{\partial \bar{f}}{\partial t} = 0$ the system is autonomous.

A non-autonomous system reduces to an autonomous by defining $\bar{y} = (\bar{x}, t) \in E \times \mathbb{R}$ so that

$$\dot{\bar{y}} = \begin{pmatrix} \bar{f}(\bar{y}) \\ 1 \end{pmatrix}.$$

Thus, can study only autonomous systems.

A nth order ODE

$$\frac{d^n x}{dt^n} = g\left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right)$$

can be reduced to a standard form with n equations by introducing

$$\bar{y} = \left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right) = (y_1, y_2, \ldots, y_n)$$

so that

$$\dot{\bar{y}} = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \\ g(y_1, y_2, \ldots, y_n) \end{pmatrix}.$$

Similarly to autonomous systems, will only consider maps of the form

$$\bar{x}_{n+1} = \bar{F}(\bar{x}_n) \qquad \text{and not} \qquad \bar{F}(\bar{x}_n, n).$$

1.2 Initial value problem (IVP)

We would like to explore an initial value problem for $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$:

> Given the initial value of $\bar{x}$: $\bar{x}(t_0) = \bar{x}_0$, $\bar{x}_0 \in E$, $t \in I \subseteq \mathbb{R}$, find a differentiable function $\bar{x}(t)$ for $t \in I$ which remains in E for $t \in I$ and satisfies the initial condition and the ODE.

For an autonomous system can define solution in terms of a flow ϕ_t.

Definition (Flow) $\phi_t(\bar{x})$ such that $\phi_t(\bar{x}_0)$ is the solution of $\dot{\bar{x}} = \bar{f}(\bar{x})$ starting at $\bar{x}_0$ when $t = 0$ is called the flow through $\bar{x}_0$ at $t = 0$.

Thus,

$$\phi_0(\bar{x}_0) = \bar{x}_0, \qquad \phi_s\big(\phi_t(\bar{x}_0)\big) = \phi_{s+t}(\bar{x}_0).$$

Does this solution exist? Is it unique?

Theorem (Cauchy–Peano)

If $\bar{f}(\bar{x}, t)$ is continuous and $|\bar{f}| < M$ in the domain

$$D : \big\{|t - t_0| < \alpha,\ |\bar{x} - \bar{x}_0| < \beta\big\}$$

then the IVP has a solution for $|t - t_0| < \min(\alpha, \beta/M)$.

But uniqueness requires stronger conditions on f.

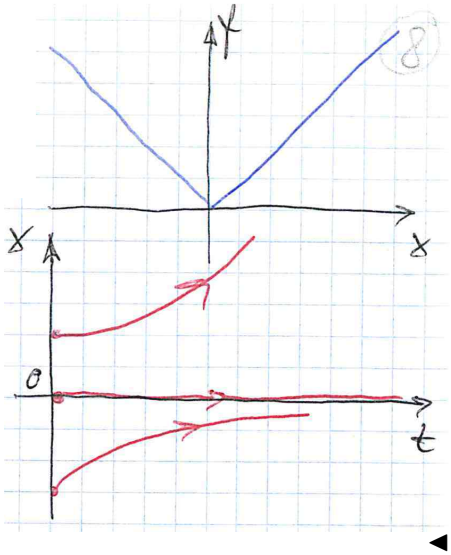
Example Unique solution

$$\dot{x} = |x|, \qquad x(t_0) = x_0$$

Solution

$$x(t) = \begin{cases} x_0 e^{t_0 - t}, & x_0 < 0, \\ 0, & x_0 = 0, \\ x_0 e^{t - t_0}, & x_0 > 0. \end{cases}$$

Note that f is continuous but not differentiable.

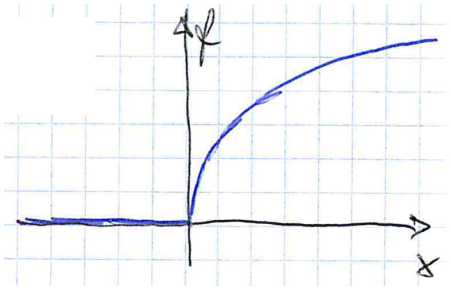


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Example Non-unique solution

$$\dot{x} = \begin{cases} \sqrt{x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad x(0) = 0$$

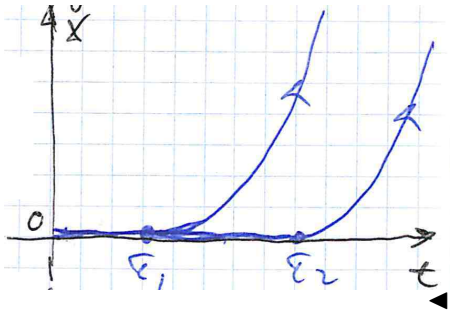
Can check that



$$x(t) = \begin{cases} 0, & t < \tau, \\ \frac{1}{4}(t - \tau)^2, & t \geq \tau \end{cases} \qquad \text{for any } \tau \geq 0$$

satisfies the equation and initial condition. But τ can be anything so a family of solutions!

Here again f is continuous and not differentiable.



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Why is the difference? Because $\dfrac{d}{dx}(\sqrt{x})$ is not bounded at 0!

To guarantee uniqueness, need f to be not just continuous, but also Lipschitz.

Definition Lipschitz property

A function $\bar{f}$ defined on a subset of $\mathbb{R}^n$ satisfies Lipschitz condition at a point $\bar{x}_0$ with Lipschitz constant L if $\exists\,(L, a)$ such that $\forall\,\bar{x}, \bar{y}$: $|\bar{x} - \bar{x}_0| < a$, $|\bar{y} - \bar{x}_0| < a$ we have

$$|\bar{f}(\bar{x}) - \bar{f}(\bar{y})| \leq L|\bar{x} - \bar{y}|.$$

Note that Differentiable $\rightarrow$ Lipschitz $\rightarrow$ Continuous

Theorem Uniqueness/Picard–Lindelöf/Cauchy–Lipschitz

Consider an IVP $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$ with $\bar{x} = \bar{x}_0$ at $t = t_0$. If $\bar{f}$ satisfies a Lipschitz condition at $\bar{x}_0$ then the solution $\phi_{t-t_0}(\bar{x}_0)$ exists and is unique and continuous in a neighborhood of $(\bar{x}_0, t_0)$.

However, in general uniqueness & continuity do not necessarily mean existence of solution for all time!

Example Finite time blowup

$$\dot{x} = x^2, \qquad x(0) = 1, \qquad \text{solution is } x(t) = \frac{1}{1-t},$$

blows up to ∞ at $t \rightarrow 1$.

In what follows, unless stated otherwise, we will consider only differentiable functions f.

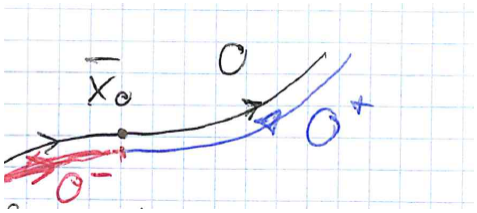
§1.3. Orbits, invariant sets and limit sets

Consider a flow $\phi_t(\bar{x}_0)$.

Definition

* The orbit/trajectory of ϕ_t through $\bar{x}_0$ is the set

$$O(\bar{x}_0) = \{\phi_t(\bar{x}_0) : -\infty < t < \infty\}.$$



* The forward orbit of ϕ_t through $\bar{x}_0$ is the set

$$O^+(\bar{x}_0) = \{\phi_t(\bar{x}_0) : 0 \le t\};$$

analogously, backward orbit $O^-(\bar{x}_0)$ if for $t \le 0$.

Definition

A set of points $\mathcal{M} \subseteq E$ is invariant under f if

$$\bar{x} \in \mathcal{M} \quad \Rightarrow \quad O(\bar{x}) \subseteq \mathcal{M}.$$

Clearly, any orbit is invariant.

Definition Fixed point

The point $\bar{x}_0$ is a fixed point (equilibrium, stationary, critical point) if $f(\bar{x}_0) = 0$. Then $\bar{x} = \bar{x}_0$ for any t and $O(\bar{x}_0) = \bar{x}_0$.

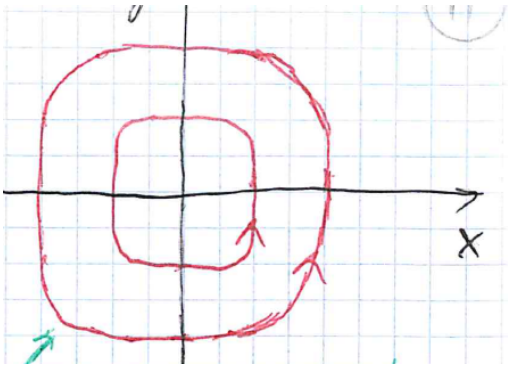
Definition Periodic point & orbit

A point $\bar{x}_0$ is a periodic point with period T if $\phi_T(\bar{x}_0) = \bar{x}_0$ for some $T > 0$ but $\phi_t(\bar{x}_0) \neq \bar{x}_0$ for $0 < t < T$.

The set $\{\phi_t(\bar{x}_0) : 0 \le t < T\}$ is the periodic orbit through $\bar{x}_0$.

Example Periodic orbit

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y^3\\ x^3\end{pmatrix} \quad \Rightarrow \quad \underline{x^4 + y^4 = \text{cst}(t)}$$



not a limit cycle!

Definition Limit cycle

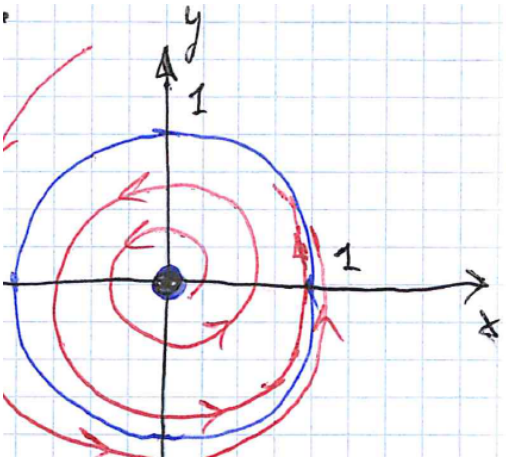
A limit cycle is an isolated periodic orbit, i.e. there are no other periodic orbits in its sufficiently small neighborhood.

Example Limit cycle

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y + x(1-x^2-y^2)\\ x + y(1-x^2-y^2)\end{pmatrix} \quad \Rightarrow$$

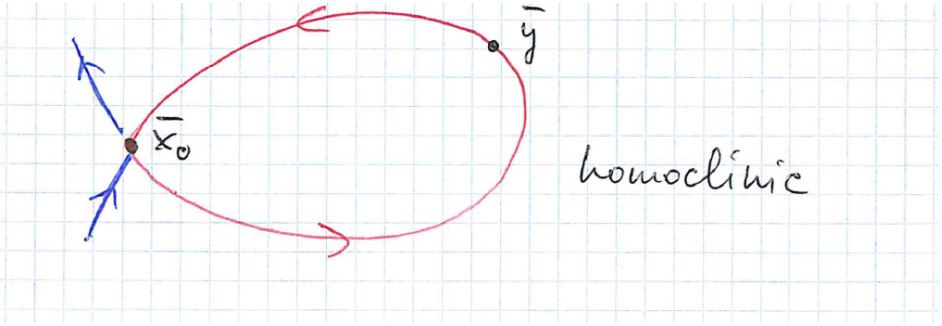
$$\underline{\dot{r} = r(1-r^2)}, \qquad \underline{\dot{\theta} = 1}.$$

A fixed point $(0,0)$ and a limit cycle $r = 1$.



Definition Homoclinic orbit

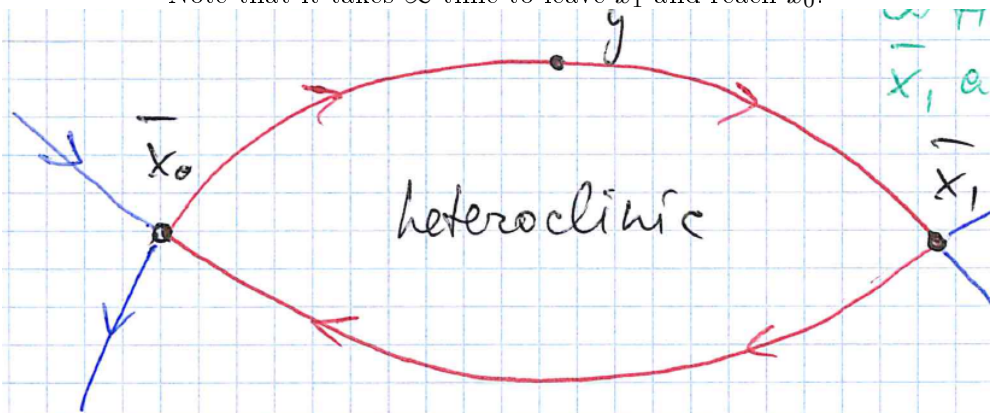
If $\bar{x}_0$ is a fixed point and $\exists\, \bar{y} \neq \bar{x}_0$ such that $\phi_t(\bar{y}) \to \bar{x}_0$ as $t \to \infty$ and $t \to -\infty$, then $O(\bar{y})$ is a homoclinic orbit.



Definition Heteroclinic orbit

If $\bar{x}_0$ and $\bar{x}_1 \neq \bar{x}_0$ are fixed points and $\exists\, \bar{y}$ such that $\phi_t(\bar{y}) \to \bar{x}_0$ as $t \to \infty$ and $\phi_t(\bar{y}) \to \bar{x}_1$ at $t \to -\infty$, then $O(\bar{y})$ is a heteroclinic orbit.

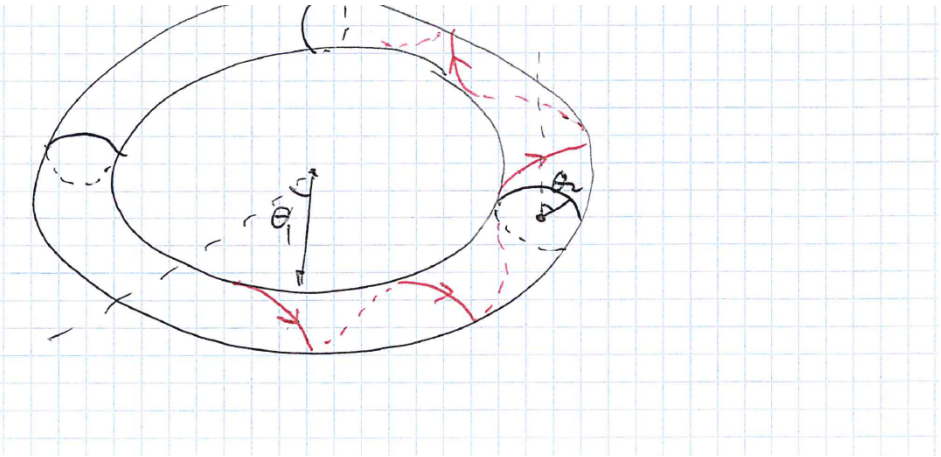
Note that it takes ∞ time to leave $\bar{x}_1$ and reach $\bar{x}_0$!



In higher dimensions more complex invariant sets are possible.

Example 2-torus in 3D

θ_1, θ_2 coordinates on the surface of a torus in 3D, $\dot{\theta}_1 = \omega_1$, $\dot{\theta}_2 = \omega_2$ and ω_1, ω_2 are not in a ratio, $\Rightarrow$ trajectory will fill the full surface of a torus. Thus, surface is an invariant set.



When talking about the long-term behavior of trajectories need to be careful with statements like $\phi_t(\bar{x}) \to \bar{y}$ as such limit may not exist, e.g. for periodic orbits.

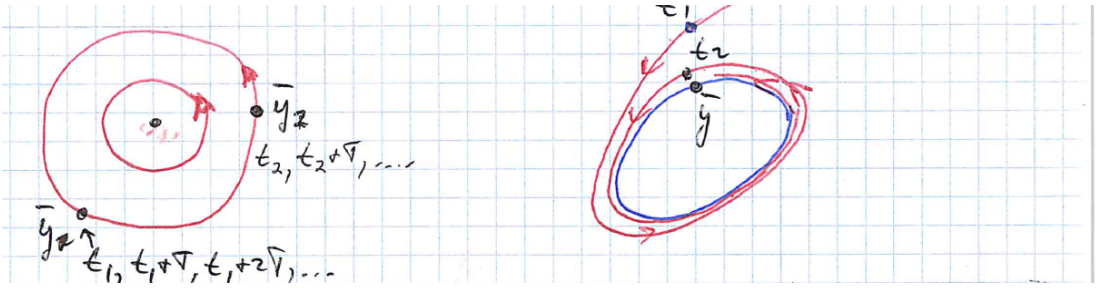
Instead use

Definition

ω-limit set of $\bar{x}$ is

$$\omega(\bar{x}) = \{\bar{y} : \exists \text{ some sequence } t_1, t_2, \ldots \to \infty \text{ with } \phi_{t_n}(\bar{x}) \to \bar{y}\}\,.$$

α-limit set defined similarly but for $t \to -\infty$.



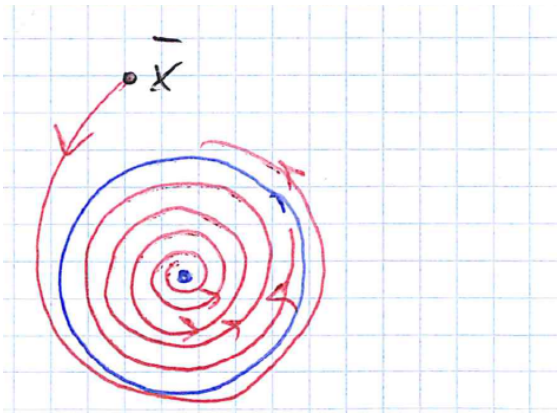
When the orbit $O(\bar{x})$ is bounded, $\omega(\bar{x})$ has properties:

* non-empty
* closed
* connected
* invariant under f
* bounded

Example

Go back to

$$\dot{r} = r(1 - r^2), \qquad \dot{\theta} = 1.$$



Here $\omega(\bar{x}) = \{r = 1\}$,

$\alpha(\bar{x}) = \{(0,0)\}$ for $|\bar{x}| < 1$ or undefined for $|\bar{x}| > 1$.

§1.4. Topological equivalence of flows

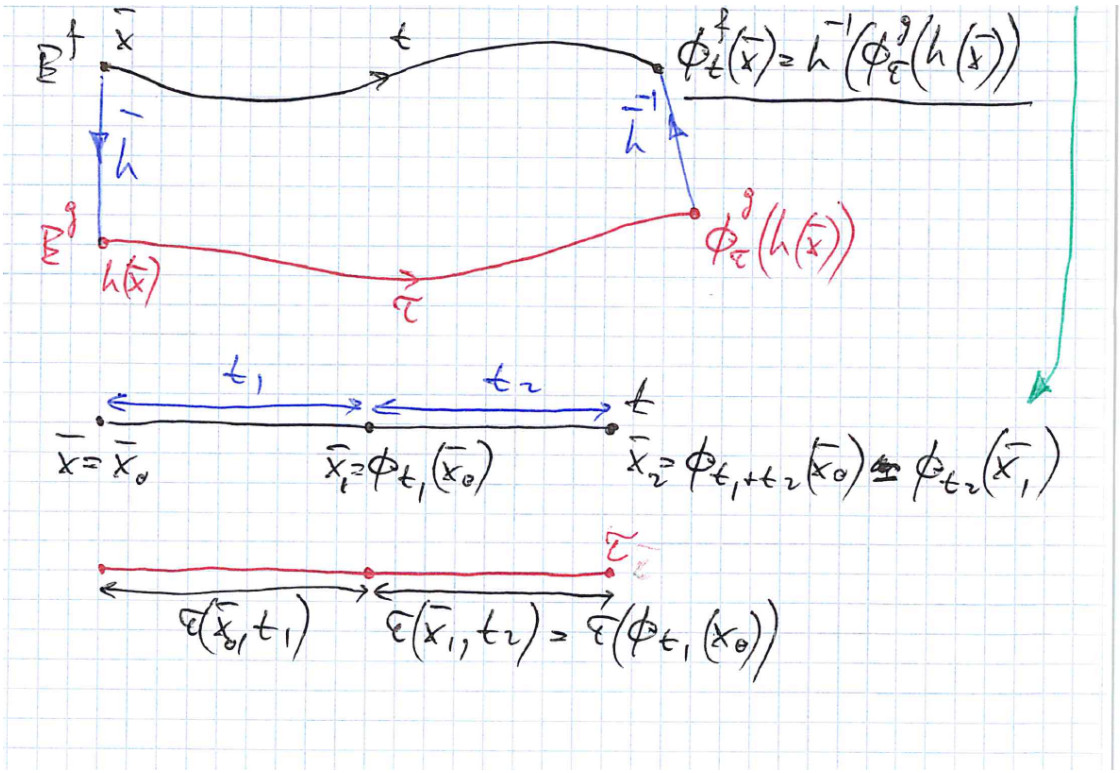
In this course we will sometimes be referring to different dynamical systems/flows as topologically/geometrically equivalent. Later we'll see that the flow structure changes at bifurcation.

Definition Topological equivalence

Two flows $\phi_t^f(\bar{x})$ and $\phi_t^g(\bar{x})$ are topologically equivalent if there is a homeomorphism $h(\bar{x}) : E^f \to E^g$ (i.e. a continuous bijection with continuous inverse) and time-increasing function $\tau(\bar{x}, t)$ such that for any $t, \bar{x}$

$$\phi_t^f(\bar{x}) = h^{-1} \circ \phi_\tau^g \circ h(\bar{x})$$

$$\text{and} \qquad \tau(\bar{x}, t_1 + t_2) = \tau(\bar{x}, t_1) + \tau\big(\phi_{t_1}^f(\bar{x}), t_2\big).$$



End of Lecture 3

§1.4. Topological equivalence of flows

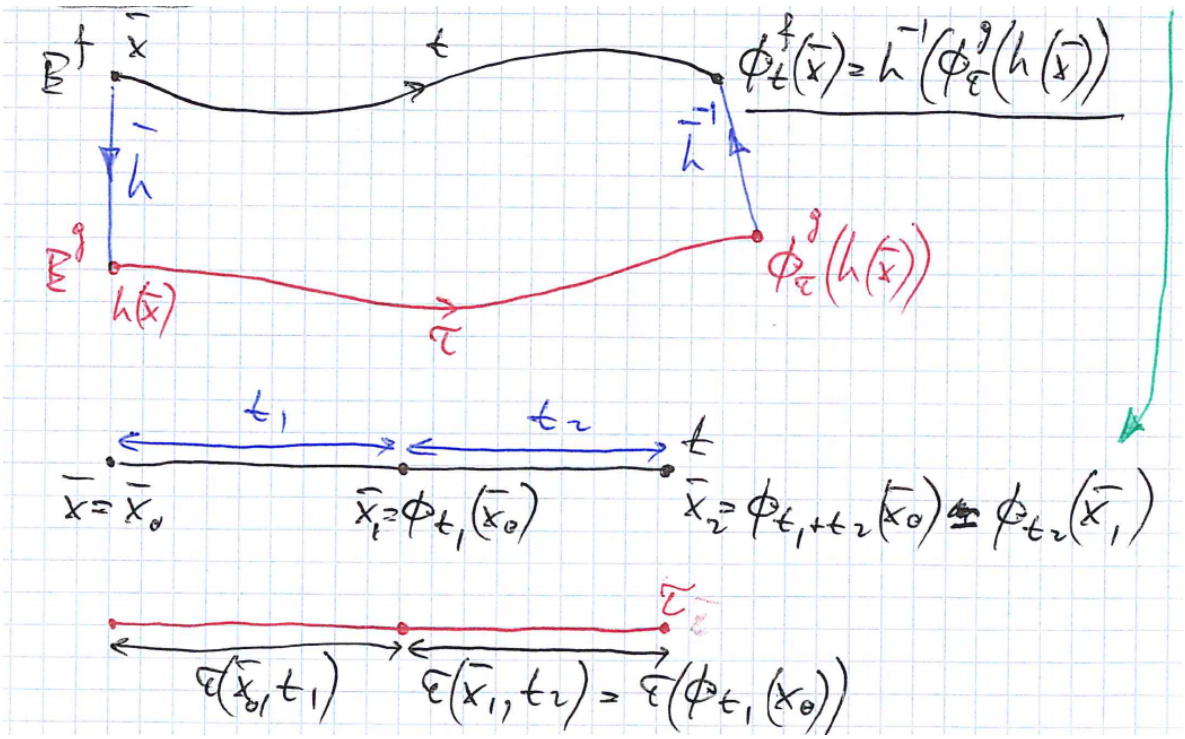
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$$\phi_t^f(\bar{x}) = \bar{h}^{-1} \circ \phi_\tau^g \circ h(\bar{x})$$

and $$\tau(\bar{x}, t_1 + t_2) = \tau(\bar{x}, t_1) + \tau\big(\phi_{t_1}^f(\bar{x}), t\big).$$



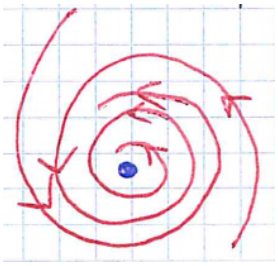
In words, $\exists$ a 'nice' mapping from one phase-space to the other and a map from one time to another such that the evolutions of the two systems are the same.

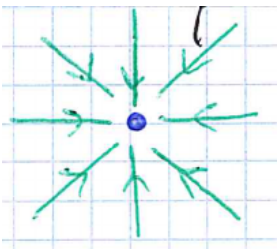
Clearly, $\bar{h}$ must map fixed points into fixed points and periodic orbits to periodic orbits (with generally different period).

If $\tau = t$ we say that the two flows are topologically conjugate.

Example

$$\begin{aligned} \dot{r} &= -r \\ \dot{\theta} &= 1 \end{aligned} \qquad \text{and} \qquad \begin{aligned} \dot{\rho} &= -2\rho \\ \dot{\psi} &= 0 \end{aligned}$$





Flows are topologically equivalent with $\bar{h}(0,0) = (0,0)$, $h(r,\theta) = (r^2, \theta + \ln r)$ for $r \neq 0$ in polars and $\tau = t$ (so topologically conjugate).

To show, solve both: starting at $\bar{x} = (r_0, \theta_0)$

$$\phi_t^f = (r_0 e^{-t}, \theta_0 + t), \qquad \phi_t^g = (\rho_0 e^{-2t}, \psi_0).$$

Then

$$h \circ \phi_t^f = \left(r_0^2 e^{-2t}, \theta_0 + t + \ln(r_0 e^{-t})\right) = \left(r_0^2 e^{-2t}, \theta_0 + \ln r_0\right)$$

and

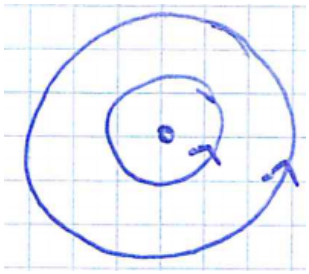
$$\phi_t^g \circ h = \left(r_0^2 e^{-2t}, \theta_0 + \ln r_0\right) \quad \text{– the same, } \Rightarrow \text{ Ok}$$

$$\text{as } h(\bar{x}) = (r_0^2, \theta_0 + \ln r_0).$$

◀

Example

$$\begin{aligned}\dot{r} &= 0 \\ \dot{\theta} &= 1\end{aligned} \qquad \text{and} \qquad \begin{aligned}\dot{r} &= 0 \\ \dot{\theta} &= r + \sin^2\theta\end{aligned}$$



Both are periodic circular orbits, so $\bar{h}(\bar{x}) = \bar{x}$, and time is stretched:

$$\frac{d\theta}{dt} * (r + \sin^2\theta) = \frac{d\theta}{d\tau}, \quad \Rightarrow$$

$$\tau = \int_0^t \frac{dt'}{r_0 + \sin^2(\theta_0 + t')}$$

as first flow is

$$\phi_t(r_0, \theta_0) = (r_0, \theta_0 + t).$$

Thus, topologically equivalent (but not conjugate). ◀

Definition

The vector field $\bar{f}$ (or system $\dot{\bar{x}} = \bar{f}(\bar{x})$ or flow $\phi_t(\bar{x})$) is structurally stable if $\exists\, \varepsilon > 0$ such that $\bar{f} + \bar{\delta}$ is topologically equivalent to $f\ \forall\, \bar{\delta}(\bar{x})$ with

$$|\bar{\delta}| + \sum_i \left| \frac{\partial \bar{\delta}}{\partial x_i} \right| < \varepsilon.$$

In the first example system is structurally stable, in the second it is not (a small $\delta \neq 0$ will turn $\dot{r} = 0$ into $\dot{r} = \delta$, which is no longer a periodic orbit).

We will discuss this later in the context of bifurcations.

§2. Behavior near fixed points

Suppose $\bar{x}_0$ is a fixed point of $\dot{\bar{x}} = \bar{f}(\bar{x})$, i.e. $\bar{f}(\bar{x}_0) = 0$. If f is smooth near $\bar{x}_0$, set

$$\bar{y} = \bar{x} - \bar{x}_0, \quad \Rightarrow \quad \dot{\bar{y}} = (\bar{x} - \dot{\bar{x}}_0) = \dot{\bar{x}} = \bar{f}(\bar{x}) = \bar{f}(\bar{y} + \bar{x}_0).$$

Taylor expand r.h.s.:

$$\dot{\bar{y}} = \underbrace{\bar{f}(\bar{x}_0)}_{=0} + \hat{A}\bar{y} + O(|y|^2), \qquad A_{ij} = \frac{\partial f_i}{\partial x_j} \quad \begin{matrix} \text{– the Jacobian} \\ \text{matrix of } f \text{ at } \bar{x}_0. \end{matrix}$$

Thus, near $\bar{x}_0$ the behavior of the system is approximated by $\dot{\bar{y}} = \hat{A}\bar{y}$.

§2.1 2D flows near fixed points

In 2D is a 2×2 matrix $\hat{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\mathrm{Det}(\hat{A}) = ad - bc$, $\mathrm{Tr}(\hat{A}) = a + d$.

Eigenvalues satisfy characteristic eq-n

$$\lambda^2 - \lambda \cdot \mathrm{Tr}(\hat{A}) + \det(\hat{A}) = 0, \quad \Rightarrow \quad \lambda = \frac{1}{2}\left[\mathrm{Tr}(\hat{A}) \pm \sqrt{(\mathrm{Tr}(\hat{A}))^2 - 4\,\mathrm{Det}(\hat{A})}\right].$$

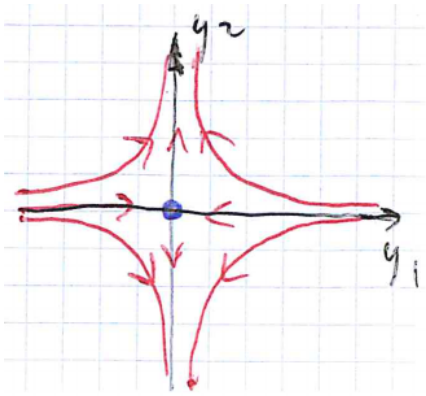
By finding eigenvectors (when we can) we can put $\hat{A}$ into canonical form & analyze.

① Saddle point: $\mathrm{Det}(\hat{A}) < 0$, $\Rightarrow \lambda_1, \lambda_2$ are real and have opposite signs, $\lambda_1\lambda_2 < 0$.

$$\hat{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \text{e.g. } \lambda_1 < 0 < \lambda_2$$

$$\Updownarrow$$

$$\dot{y}_1 = \lambda_1 y_1, \quad \dot{y}_2 = \lambda_2 y_2, \quad \Rightarrow \quad \begin{matrix} y_1 = y_{1,0} e^{\lambda_1 t} \\ y_2 = y_{2,0} e^{\lambda_2 t} \end{matrix}$$



End of Lecture 4

Since, eliminating t, $y_2 \propto y_1^{\lambda_2/\lambda_1} = \dfrac{1}{y_1^{|\lambda_2/\lambda_1|}}$. So, orbits are hyperbolas!

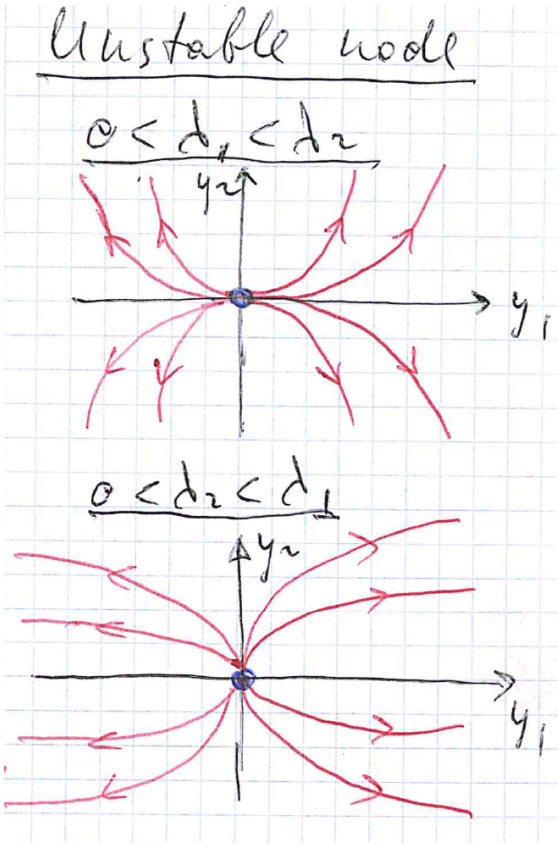
② Node: $(\mathrm{Tr}(\hat{A}))^2 > 4\,\mathrm{Det}(\hat{A}) > 0$, $\Rightarrow \lambda_1, \lambda_2$ are real and either both positive ($\mathrm{Tr}(\hat{A}) > 0$) or negative ($\mathrm{Tr}(\hat{A}) < 0$).

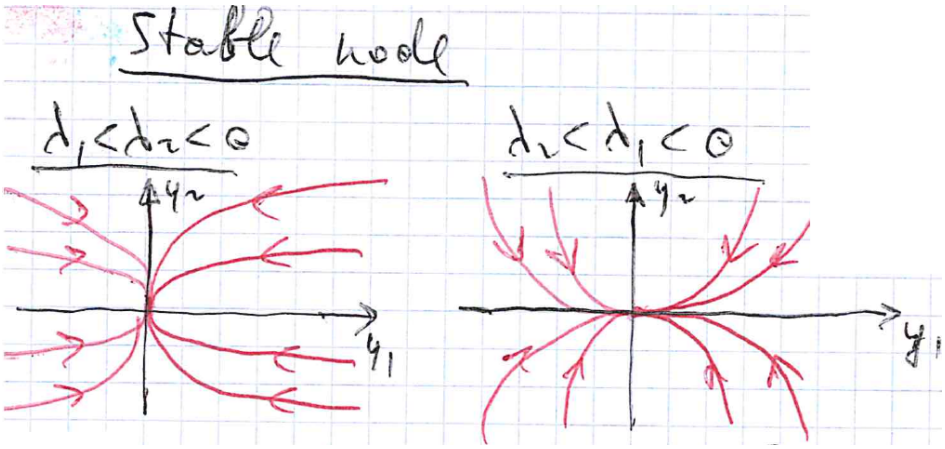
Same canonical form

$$\hat{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \Rightarrow \quad y_2 \propto y_1^{\lambda_2/\lambda_1}.$$

Get different patterns depending on $\lambda_2/\lambda_1 < 1$ or > 1.

Sign of λ_1, λ_2 sets the direction of the flow.





③ Focus (spiral): $(\mathrm{Tr}(\hat{A}))^2 < 4\,\mathrm{Det}(\hat{A})$ and $\det(\hat{A}) > 0$ so that roots are complex, $\lambda = \rho \pm i\omega$, $\omega \neq 0$.

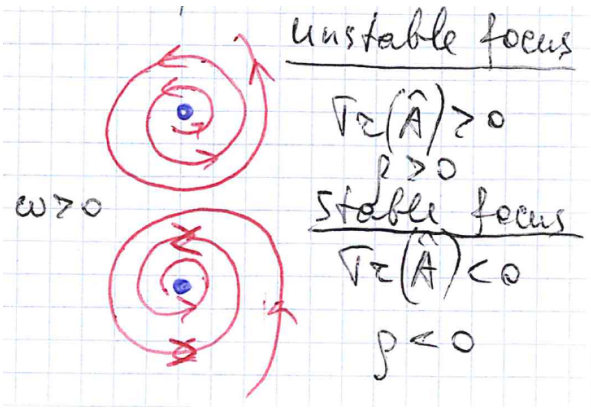
Canonical form

$$\hat{A} = \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix}.$$

In polars have

$$\dot{r} = \rho r, \quad \dot{\theta} = \omega, \quad \Rightarrow$$

$$r = r_0 e^{\rho t}, \quad \theta = \theta_0 + \omega t.$$



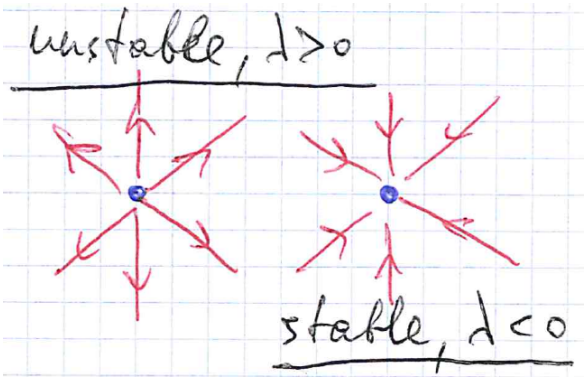
Other cases are degenerate.

④ $(\mathrm{Tr}(\hat{A}))^2 = 4\,\mathrm{Det}(\hat{A}) > 0$

A boundary between focus & node, double eigenvalue $\lambda = \frac{1}{2}\mathrm{Tr}(\hat{A})$. Two cases:

Stellar node (star)

if $\hat{A} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$ is canonical



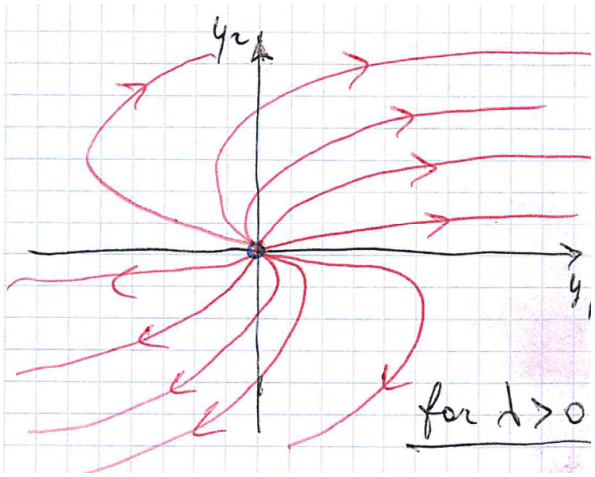
Improper node

if $\hat{A} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ is canonical

(defective)

Here $y = y_0 e^{\lambda t}$,

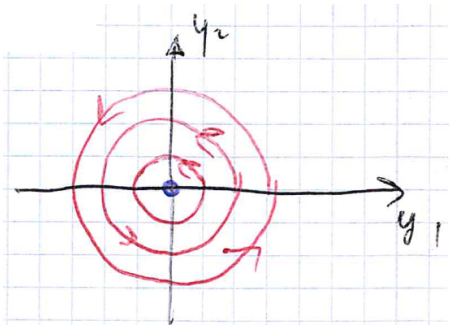
$x = x_0 e^{\lambda t} + y_0 t e^{\lambda t}$, $\quad \dfrac{y}{x} = \dfrac{y_0}{x_0 + y_0 t}$.



⑤ $\mathrm{Det}(\hat{A}) > 0$, $\quad \mathrm{Tr}(\hat{A}) = 0$: Centre

Here $\lambda = \pm i\omega$, $\qquad \hat{A} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$.

$\dot{r} = 0$, $\quad \dot{\theta} = \omega$, $\quad \Rightarrow$ orbits are closed circles.

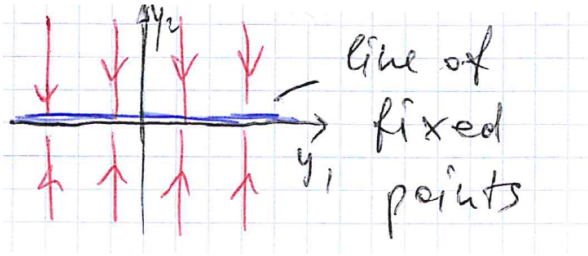


⑥ $\mathrm{Det}(\hat{A}) = 0$, $\quad \mathrm{Tr}(\hat{A}) \neq 0$ – between saddle & node

$\lambda_1 = 0$, $\quad \lambda_2 \neq 0$, $\qquad \hat{A} = \begin{pmatrix} 0 & 0 \\ 0 & \lambda \end{pmatrix}$

$y_1 = \text{const}$, $\qquad y_2 = y_{2,0} e^{\lambda t}$

for $\lambda < 0$

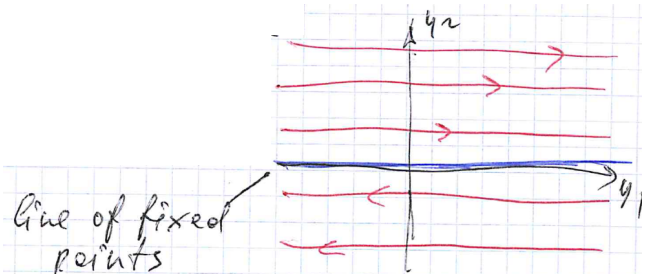


⑦ $\underline{\mathrm{Det}(\hat{A}) = 0, \quad \mathrm{Tr}(\hat{A}) = 0}$, $\lambda_{1,2} = 0$

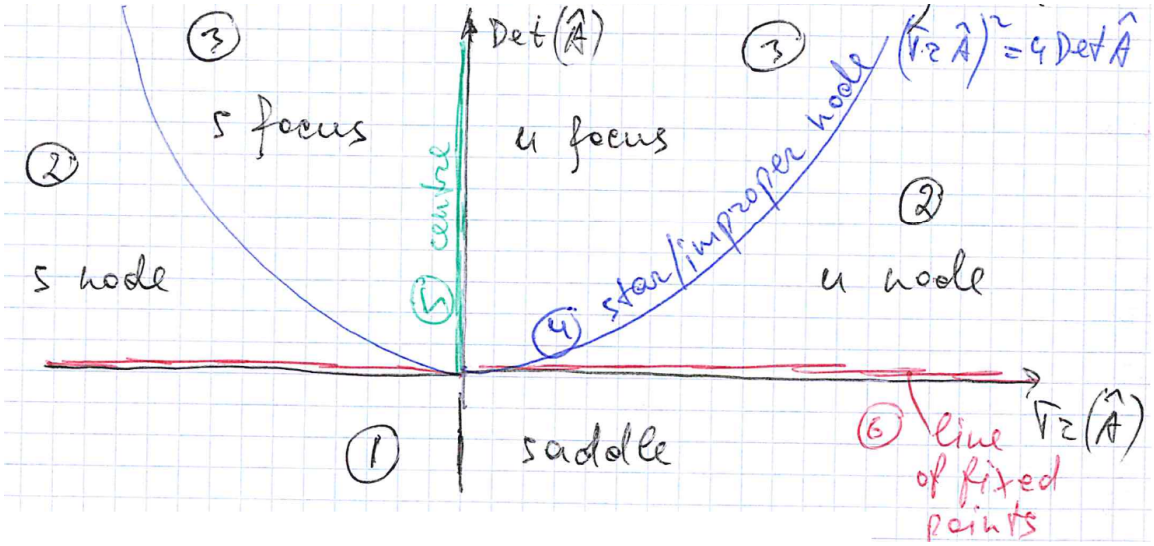
if $\hat{A} = 0$, $\Rightarrow y_1 = \mathrm{const}$, $y_2 = \mathrm{const}$, $\Rightarrow$ plane of fixed points!

if $\hat{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, this is a case of shear,

$$\underline{\dot{y}_1 = y_2, \quad \dot{y}_2 = 0.}$$



This zoo can be summarized in a diagram



Canonical forms are in special basis

$$(*) \quad \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \text{– eigenvectors}$$

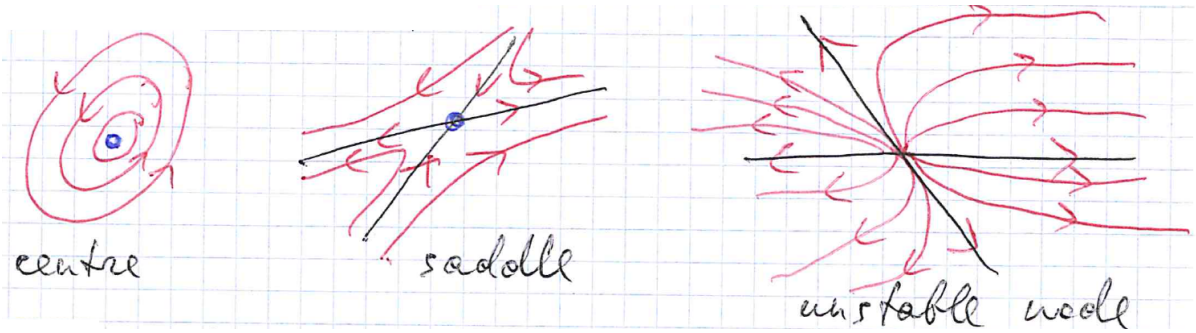
$$(*) \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \quad \text{– generalized eigenvectors}$$

$$(*) \quad \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix} \quad \text{– Re \& Im of eigenvectors}$$

Notes

(∗) No need to change basis in practice, just work with $\mathrm{Tr}(\hat{A})$, $\mathrm{Det}(\hat{A})$ to classify

(∗) In original (non-canonical) coordinates shapes can be distorted



(∗) Useful to know eigenvectors for saddles/nodes; for focus/centre cases ratios depend also on $\lambda_{1,2}$, easier to just plot several values

(∗) Centres are rare but generic in Hamiltonian systems:

$$\dot{\bar{x}} = \begin{pmatrix} \dfrac{\partial H}{\partial y} \\ -\dfrac{\partial H}{\partial x} \end{pmatrix} \quad \text{for some } H(x,y), \quad \bar{x} = (x,y).$$

$$\text{Indeed,} \quad \hat{A} = \begin{pmatrix} H_{xy} & H_{yy} \\ -H_{xx} & -H_{xy} \end{pmatrix}, \quad \Rightarrow \quad \underline{\mathrm{Tr}(\hat{A}) = 0}, \quad \Rightarrow$$

$$\text{either} \quad \underset{\lambda = \pm i\omega}{\text{centres}} \quad \text{or} \quad \underset{\lambda_1 \lambda_2 < 0}{\text{saddles}}.$$

$$\text{Also,} \quad \underline{\frac{dH}{dt}} = (\dot{\bar{x}} \cdot \nabla) H = \dot{x} H_x + \dot{y} H_y = H_y H_x - H_x H_y = 0.$$

Thus, $H(x,y) = \text{const}$ on trajectories, orbits follow contours of $H(x,y)$ in autonomous Hamiltonian systems.

End of Lecture 5

§2.2 Effect of nonlinear terms

How does the behavior of linearization $\dot{\bar{y}} = \hat{A}\bar{y}$ of the system $\dot{\bar{x}} = \bar{f}(\bar{x})$ about the fixed point $\bar{x}_0$ inform us about the behavior of the full nonlinear system around $\bar{x}_0$?

Definition

A fixed point $\bar{x}_0$ is hyperbolic if none of the eigenvalues of the Jacobian of $\bar{f}(\bar{x})$ (matrix $\hat{A}$) at $\bar{x}_0$ have zero real part. It is non-hyperbolic otherwise.

We classify hyperbolic fixed points as

- $(*)$ source if all $\mathrm{Re}(\lambda_i) > 0$
- $(*)$ sink if all $\mathrm{Re}(\lambda_i) < 0$
- $(*)$ saddle otherwise (i.e. some $\mathrm{Re}(\lambda_i) > 0$, others < 0).

We will discuss separately hyperbolic & non-hyperbolic fixed points.

One can show (e.g. see Glendinning, Ch. 4) that if

(1) $\bar{x}_0$ is hyperbolic and

(2) the nonlinear corrections of $\bar{f}$ are $O(|\bar{x} - \bar{x}_0|^2)$

then the full nonlinear system and the linearized system are topologically conjugate.

Note: non-hyperbolic fixed points arise when

$$\{\mathrm{Det}\,\hat{A} \geq 0,\ \mathrm{Tr}\,\hat{A} = 0\}, \quad \{\mathrm{Det}\,\hat{A} = 0,\ \mathrm{Tr}\,\hat{A} \neq 0\},$$

i.e. cases 5, 6, 7 identified earlier.

§2.2.1 Stable and unstable manifolds

For a linearized system we can separate the phase space into different domains according to different behaviors in time.

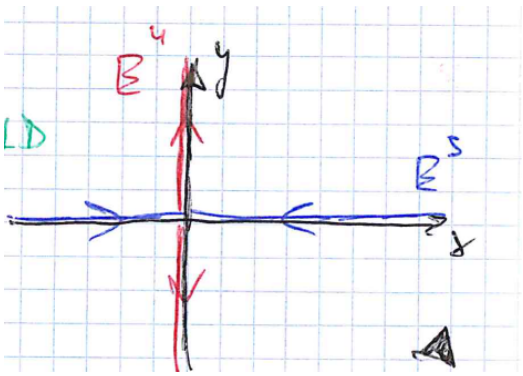
Definition

The stable, unstable and centre subspaces of the linearization of $\bar{f}(\bar{x})$ near a fixed point $\bar{x}_0$ are the three linear subspaces E^s, E^u and E^c spanned by subsets of (possibly generalized) eigenvectors of $\hat{A}$ whose eigenvalues have real parts $\mathrm{Re}(\lambda) < 0$, $\mathrm{Re}(\lambda) > 0$ and $\mathrm{Re}(\lambda) = 0$, respectively.

Note: a hyperbolic fixed point has no centre subspace.

Example

$$\underset{\text{2D}}{\hat{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}}, \quad \Rightarrow \quad \begin{matrix} E^s = x\text{-axis} \\ E^u = y\text{-axis} \end{matrix} \quad \text{1D}$$



Theorem Stable manifold theorem

Suppose $\bar{0}$ is a hyperbolic fixed point of $\dot{\bar{x}} = \bar{f}(\bar{x})$ and E^s and E^u are the stable and unstable subspaces of the linearization of $\bar{f}$ near $\bar{0}$. Then $\exists$ stable and unstable manifolds W^s_{loc} & W^u_{loc}, which have the same dimension as E^s, E^u and are tangent to E^s & E^u at $\bar{0}$, such that for $\bar{x} \neq 0$ but in a sufficiently small neighborhood of $\bar{0}$,

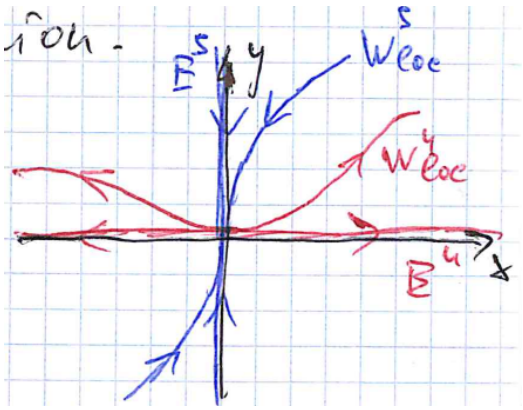
$$W^s_{\text{loc}} = \{\bar{x} : \phi_t(\bar{x}) \to \bar{0} \text{ as } t \to +\infty\},$$

$$W^u_{\text{loc}} = \{\bar{x} : \phi_t(\bar{x}) \to \bar{0} \text{ as } t \to -\infty\}.$$

Proof: not part of the course, see Glendinning.

The local stable manifold W^s_{loc} can be extended to a global invariant manifold W^s by following the flow backwards in time starting at $\bar{x} \in W^s_{\text{loc}}$. Same for W^u_{loc}, extend to W^u by $t \to +\infty$.

In $\mathbb{R}^2$, this theorem guarantees the existence of two trajectories that approach and leave a saddle fixed point in a tangent fashion.



It is often easy in practice to find local approximations to stable and unstable manifolds.

Indeed, for a saddle in $\mathbb{R}^2$ assume that E^s is $x = 0$ and E^u is $y = 0$ (can always change the basis).

Then look for W^s_{loc} in the form $x = S(y)$ with $S(0) = 0$ and $S'(0) = 0$. For W^u_{loc} take $y = U(x)$ with $U(0) = 0$, $U'(0) = 0$. These manifolds must satisfy $\dot{\bar{x}} = \bar{f}(\bar{x})$, i.e. $\dot{x} = \frac{\partial S}{\partial y}\dot{y}$ and $\dot{y} = \frac{\partial U}{\partial x}\dot{x}$ must solve $\dot{\bar{x}} = \bar{f}(\bar{x})$. Thus, for

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix},$$

$$\dot{x} = \frac{dS}{dy}\dot{y} = \underbrace{\frac{dS}{dy} g(S(y),y) = f(S(y),y)}_{\text{(nonlinear) ODE for } S(y)} \quad \text{and}$$

$$\dot{y} = \frac{dU}{dx}\dot{x} = \underbrace{\frac{dU}{dx} f(x,U(x)) = g(x,U(x))}_{\text{ODE for } U(x)}.$$

No need to solve these ODEs, we can simply find a locally convergent series expansion in the form

$$S(y) = a_2 y^2 + a_3 y^3 + \cdots, \qquad U(x) = b_2 x^2 + b_3 x^3 + \cdots$$

and solve term by term.

Example

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} x(1-y) \\ -y + x^2 \end{pmatrix}, \qquad \hat{A} = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}\Big|_{(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$\det \hat{A} = -1 \;\Rightarrow\; \underline{\text{saddle}}, \qquad \Rightarrow \quad E^s = \{x = 0\}, \quad E^u = \{y = 0\}.$$

For W^u_{loc} take

$$y = U(x) = a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots,$$

substitute into

$$\dot{y} = -y + x^2 = \frac{dU}{dx} x(1-y), \quad \Rightarrow$$

$$-U(x) + x^2 = \frac{dU}{dx} x(1 - U(x)), \quad \Rightarrow$$

$$x^2 - a_2 x^2 - a_3 x^3 - a_4 x^4 + \cdots = (2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots)\, x \times (1 - a_2 x^2 - a_3 x^3 - a_4 x^4 + \cdots), \quad \Rightarrow$$

$$\begin{aligned} x^2: &\quad 1 - a_2 = 2a_2 &&\Rightarrow \quad a_2 = \frac{1}{3} \\ x^3: &\quad -a_3 = 3a_3 &&\Rightarrow \quad a_3 = 0 \\ x^4: &\quad -a_4 = 4a_4 - 2a_2^2 &&\Rightarrow \quad a_4 = \frac{2}{45} \end{aligned}$$

due to $x \mapsto -x$ symmetry! $\quad (a_3 = 0)$

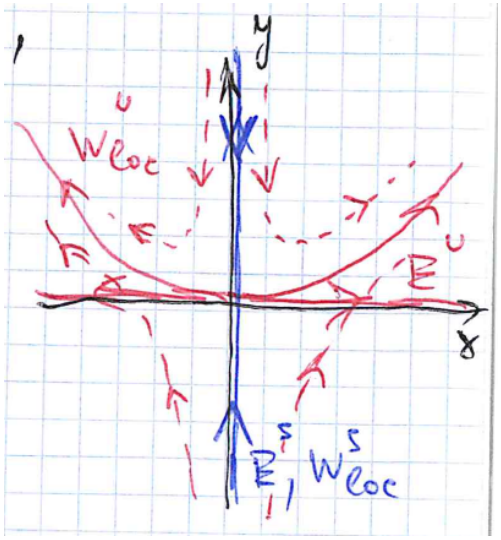
$$\text{Thus,} \quad U(x) = \frac{x^2}{3} + \frac{2}{45} x^4 + \cdots.$$

To find W^s_{loc}, $x = S(y) = a_2 y^2 + a_3 y^3 + \cdots$,

$$S(y)(1-y) = \frac{dS}{dy}[-y + S^2(y)],$$

repeat calculation, find $a_n = 0\ \forall n$. This is caused by symmetry! If $X = -x$ then system the same so W^s_{loc} has to be symmetric w.r.t. y-axis & tangent to it,

$$\Rightarrow \quad W^s_{\text{loc}} = y\text{-axis}.$$



End of Lecture 6

Note:

- If E^s & E^u are not aligned with the coordinate axes, one can either transform the coordinates to get alignment, or consider functions S & U tangent to the relevant eigenvector. E.g. for $\bar{e} = \binom{1}{k}$ use $y = kx + a_2x^2 + a_3x^3 + \cdots$.
- In higher dimensions use multi-coordinate versions of S & U. E.g. for a 2D E^s in 3D use S: $z = S(x, y)$ and $\dot{z} = \dot{x}\dfrac{\partial S}{\partial x} + \dot{y}\dfrac{\partial S}{\partial y}$ and expand in both coordinates x, y.

§2.2.2 Non-linear terms in non-hyperbolic cases

Suppose at least one eigenvalue has $\mathrm{Re}(\lambda) = 0$. Have many possibilities.

(a) $\lambda = \pm i\omega$, linear system has a centre. In nonlinear case multiple possible outcomes, e.g.

* $\begin{aligned}\dot{x} &= -y - x^3,\\ \dot{y} &= x - y^3,\end{aligned}$ $\Rightarrow \quad \dot{r} = -\dfrac{x^4 + y^4}{r} \le 0,$

$$\dot{\theta} = 1 - \frac{xy^3 - yx^3}{r^2} \to 1 \quad \text{as } r \to 0.$$



$H = -\frac{1}{2}(x^2 + y^2)$ for linear.

* $\begin{aligned}\dot{x} &= -y + x^3,\\ \dot{y} &= x + y^3,\end{aligned}$ $\Rightarrow \quad \dot{r} = \dfrac{x^4 + y^4}{r} \ge 0,$

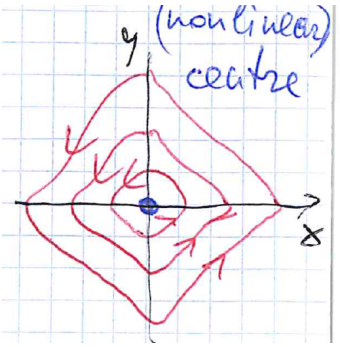
$$\dot{\theta} \to 1 \quad \text{as } r \to 0.$$



* $\begin{aligned}\dot{x} &= -y - 2x^2y,\\ \dot{y} &= x + 2y^2x\end{aligned}$ Hamiltonian system,

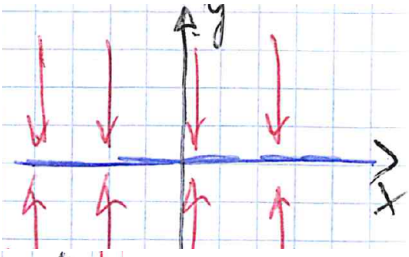
$$\dot{x} = \frac{\partial H}{\partial y}, \qquad \dot{y} = -\frac{\partial H}{\partial x},$$

with $H = -\frac{1}{2}(x^2 + y^2) - x^2y^2$ — contours are orbits.



(b) $\lambda_1 = 0$, $\lambda_2 \neq 0$, linearized — line of fixed points.

Some possibilities:



* $\begin{aligned} \dot{x} &= x^2, \\ \dot{y} &= -y, \end{aligned}$ linearized $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$,

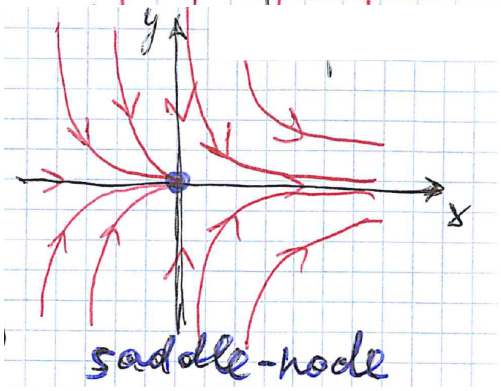
$$\text{Det}\,\hat{A} = 0, \quad \text{Tr}\,\hat{A} = -1.$$

Full solution:

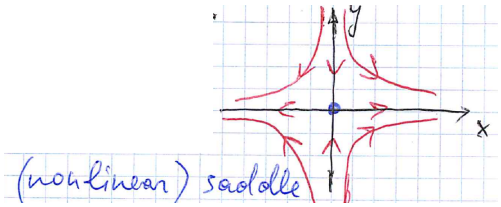
$$y \propto \exp\left(\frac{1}{x}\right), \quad \dot{x} > 0, \quad y \propto e^{-t}.$$

Note: $E^c = \{y = 0\}$.

In the nonlinear case have multiple (non-unique) manifolds tangent to E^c at $(0,0)$ — no analogue of stable manifold theorem for centre manifolds.

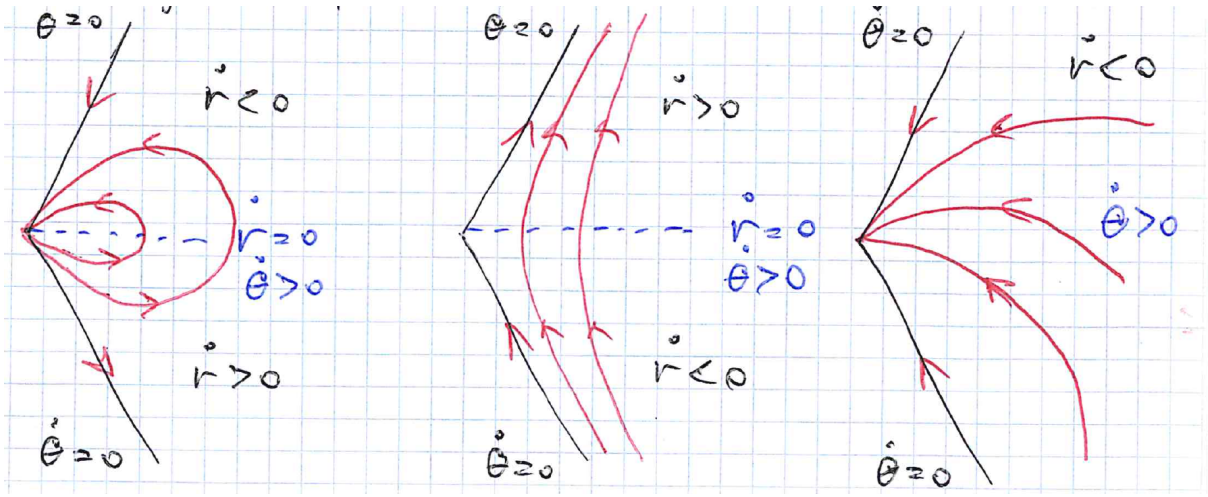


* $\begin{aligned} \dot{x} &= x^3, \\ \dot{y} &= -y, \end{aligned}$ $y \propto \exp\left(\frac{1}{2x^2}\right)$.



(c) $\lambda_1 = \lambda_2 = 0$, linearized — plane or line of fixed points.

Almost anything is possible! Change to polars, find rays $\dot{\theta} = 0$, between them can have



§2.3 Sketching phase portraits

Example $\dot{x} = x(1-y),$ $\dot{y} = -y + x^2,$ $x \mapsto -x$ symmetry, $\Rightarrow$ only $x \geq 0$ is enough.

Fixed points $(0,0)$, $(1,1)$, $(-1,1)$. FP at $(0,0)$ is a saddle, studied before. Focus on $(1,1)$.

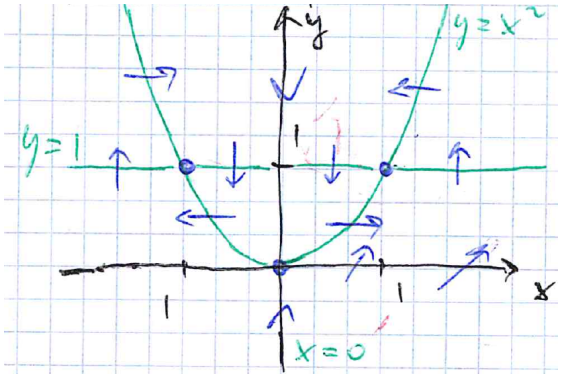
Linearize:

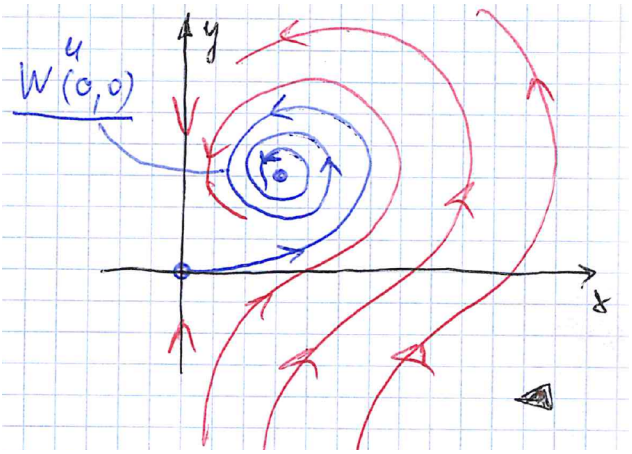
$$\hat{A} = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}, \qquad \hat{A}_{(1,1)} = \begin{pmatrix} 0 & -1 \\ 2 & -1 \end{pmatrix},$$

$$\operatorname{Tr}\hat{A} = -1, \quad \operatorname{Det}\hat{A} = 2, \quad (\operatorname{Tr}\hat{A})^2 < 4\operatorname{Det}\hat{A} \quad \Rightarrow \quad \text{focus},$$

$$\lambda = \rho \pm i\omega \quad \text{and} \quad \rho = \tfrac{1}{2}\operatorname{Tr}\hat{A} < 0 \quad \Rightarrow \quad \text{stable focus}.$$

Nullclines (where $\dot{x} = 0$, $\dot{y} = 0$ separately) get signs of $\dot{y}$, $\dot{x}$ on them.





General strategy

(1) Check for symmetries, find fixed points

(2) Calculate Jacobian, classify fixed points

(3) For saddles, find eigenvectors of Jacobian

(4) Find nullclines (where $\dot{x}$ or $\dot{y} = 0$) and determine flow direction on them and around

(5) Use knowledge of stable/unstable manifolds

(6) Combine these features into a global picture

(7) Trajectories cannot cross!

(8) Search for periodic orbits (see Chapter 4)

End of Lecture 7